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A hybrid deep learning model for water quality prediction

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ABSTRACT Water quality prediction is crucial for water environment management, playing a key role in preventing pollution and ensuring water resource safety. To improve prediction accuracy, this paper proposes a CEEMDAN-CNN-LSTM model enhanced by a self-attention mechanism. First, the Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) algorithm decomposes the water quality time series into multiple Intrinsic Mode Functions (IMFs), reducing noise and addressing nonlinearities in the data. Next, Convolutional Neural Networks (CNN) extract spatial features from the decomposed IMFs, improving the model's spatial learning capability. Then, a Long Short-Term Memory (LSTM) network models the temporal features, capturing long-term dependencies in the water quality data. Finally, a self-attention mechanism assigns varying importance to features at different time steps, improving the model's focus on key information. Experimental results demonstrate that the proposed model outperforms traditional methods, achieving high accuracy across multiple water quality indicators. This model effectively addresses non-stationarity and noise in the data, demonstrating strong generalization and robustness, providing a novel and efficient approach to water quality prediction.

INDEX TERMS Water quality prediction, CEEMDAN, Deep learning, Self-attention

I. INTRODUCTION

WITH the advancement of global industrialization, water pollution has become increasingly serious, posing a significant threat to both the ecological environment and human health. As an indispensable and invaluable resource on Earth, a decline in water quality can lead to widespread health risks and ecological instability [1]. Consequently, water quality monitoring and early warning systems have become critical components of environmental protection and public safety. In this context, research on water quality prediction has garnered increasing attention. Accurate water quality prediction not only provides a solid scientific foundation for water resource management but also enables timely warnings of potential pollution events, thereby improving response efficiency to water contamination.

The core of water quality prediction lies in utilizing existing water quality monitoring data to accurately forecast future changes. However, the multivariate, nonlinear, and time-varying nature of water quality data presents significant challenges for prediction. Traditional water quality prediction methods, such as linear regression, time series

analysis, and statistical techniques, struggle to effectively capture nonlinear relationships and long-term dependencies in complex water quality data [2]. In recent years, the rapid development of machine learning, particularly deep learning, has offered new solutions for water quality prediction. Deep learning, with its ability to extract features and recognize patterns from complex data, has shown significant advantages in handling high-dimensional and nonlinear datasets [3]. As a result, deep learning based water quality prediction methods have become an emerging research focus. Currently, the deep learning models most commonly applied to water quality prediction include Convolutional Neural Networks (CNN) [4], Long Short-Term Memory Networks (LSTM) [5], and their hybrid models [6] [7]. Each of these models has distinct characteristics: CNN is particularly effective at extracting local features and is well-suited for capturing spatial patterns in water quality data; LSTM can effectively handle the long-term dependencies of well in modeling the temporal relationships of water quality data. However, these single models have some limitations when applied to complex water quality prediction, such

as inability to effectively capture long-term dependencies or lack of robustness in handling time-varying data. In recent years, more and more researchers have attempted to combine multiple deep learning models to improve the accuracy of water quality prediction by constructing hybrid models.

This paper proposes a CEEMDAN-LSTM-CNN hybrid prediction model based on the self-attention mechanism with the aim of enhancing water quality prediction performance. The model achieves this goal through a four-stage collaborative optimization process. First, the CEEMDAN algorithm is used to decompose the original water quality time-series data at multiple scales, generating intrinsic mode functions (IMFs) that represent different frequency features, effectively separating high-frequency fluctuations, low-frequency trends, and residual components, thereby improving the interpretability of the data. Second, the decomposed IMFs were input into the LSTM network to model the long-term dynamic evolution of each component. Subsequently, a one-dimensional CNN was used to thoroughly analyze the LSTM output, uncovering local spatial patterns and high-level abstract features. Finally, a self-attention mechanism was introduced to dynamically allocate feature weights and enhance the sensitivity of the model to key time steps. Compared to existing methods, this study is innovative in three aspects: First, by cascading CEEMDAN with LSTM, the model prioritizes the decomposition of multi-scale time-series features, providing structured input for the subsequent model; second, it overturns the traditional "CNN followed by LSTM" architecture by adopting the reverse design of LSTM-CNN, achieving deep coupling of temporal dependency modeling and spatial feature extraction; and third, it constructs a multi-scale feature dynamic weighting framework, combining the self-attention mechanism to effectively suppress noise interference, thereby improving the model's generalization capability for non-stationary water quality data.

The significance of this research lies in its potential to support real-time decision making in smart water management systems. Accurate water quality prediction can not only reduce the economic losses caused by sudden pollution events, but also provide scientific evidence for watershed ecological restoration and drinking water safety assurance. In particular, with increasing water quality fluctuations due to climate change, developing high-precision and robust prediction models is of urgent practical importance. The core goal of this study is to explore performance improvement paths for existing hybrid models in multi-scale feature extraction, long-term dependency modeling, and dynamic feature selection through algorithmic innovation and architectural optimization and to provide new research ideas for intelligent prediction methods in complex water quality scenarios. Future experiments will verify the model's superiority in multiple indicators (such as dissolved oxygen and turbidity) and different prediction periods, as well as the synergistic effects of CEEMDAN decomposition and

the self-attention mechanism on performance enhancement, offering feasible technical references for the optimization of water quality monitoring systems in practical engineering.

II. RELATED WORK

Generally, water quality prediction methods can be divided into physically-driven methods and data-driven methods [8]. The physical driven method is based on various data to establish a hydrodynamic water quality model, which simulates and predicts the water quality changes of the cross-section based on the current state and the upstream and downstream boundaries of the future period. However, such methods require consideration of a large number of data sources, resulting in overly complex models. The data-driven method is generally based on long series historical monitoring data and uses machine learning algorithms to predict water quality in the future. The basic principle and implementation process of this kind of method are simple, so it is widely used. Common data-driven methods include time series models [9] [10] [11], support vector machine [12] [13] and neural network [14] [15] [16] [17] [18].

The time-series prediction method is based on water quality data, using methods such as the weighted average to establish corresponding mathematical models, and inferring future changes in water quality indicators from historical data. This method is often used for medium to short term water quality prediction. Liu et al. [9] proposed a water quality autoregression prediction model based on Markov-switching theory, and the model is improved by introducing the crow search algorithm to obtain optimal model parameters. The model has a significant predictive effect on the water quality parameter CODMn. Hirsch et al. [10] developed a weighted regression model based on time, discharge, and season (WRTDS) for river water quality trend research. Zhang et al. [11] improved WRTDS estimation by accounting for the autocorrelation structure of model residuals using the first-order autoregressive model. The experimental results show that the improved method has obvious effect on the estimation of nitrate-plus-nitrite (NOx), total phosphorus, total Kjeldahl nitrogen, soluble reactive phosphorus, suspended sediment, and chloride. In practical operation, this type of method is greatly affected by sampling frequency, making it difficult to achieve the best prediction accuracy.

Support vector machine (SVM) is one of the most influential machine learning algorithms in Supervised learning. Some scholars have attempted to use SVM for water quality prediction and achieved satisfactory results. Yahya et al. [12] developed an efficient model using SVM to predict the water quality through the analysis of the data of six parameters of dual reservoirs that are located in the catchment. This model can accurately predict water quality factors with relatively small errors, and can also help optimize water quality monitoring plans by decreasing the frequency and quantity of sampling sites, as well as the number of water quality factors. Y et al. [13] proposed a prediction method based

Method Type	Specific Model	Key Findings
Time Series Models	Autoregressive Model (Markov-switching)	Introduced crow search algorithm to optimize parameters, significantly improving COD _{Mn} prediction accuracy.
	Weighted Regression Model (WRTDS)	Developed a weighted regression model based on time, discharge, and season for water quality trend analysis.
	Improved WRTDS Model	Corrected residual autocorrelation using first-order autoregressive model, enhancing multi-parameter prediction accuracy.
Support Vector Machine	Supervised Learning Model	Optimized monitoring plans by reducing sampling frequency and sites with small errors.
	Least Squares SVM Regression	Combined with natural heuristic algorithm to improve dissolved oxygen prediction accuracy.
Neural Networks	Feedforward Neural Network (FNN)	Backpropagation neural networks effectively predict dissolved oxygen.
	Backpropagation NN (BPNN)	Improved model accuracy using modified bee colony algorithm.
	Long Short-Term Memory (LSTM)	Utilized Pearson correlation for feature extraction but required high computational costs.
	LSTM Deep Neural Network	Standardized procedure for automated training across diverse datasets.
	Bi-LSTM + Attention Mechanism	Highlighted critical data sequences via attention mechanism, improving prediction performance.
	LSTM Seq2Seq + XGBoost + KNN	Reduced RMSE by 12% (short-term) and MAE by 50% (long-term), lowering false alarm rates.

TABLE 1: Summary of Water Quality Prediction Methods

on natural heuristic algorithm and least squares support vector regression model for predicting dissolved oxygen concentration. Generally speaking, SVM are suitable for predicting small datasets, but their predictive advantages are not significant in larger datasets.

Neural networks have strong nonlinear fitting ability and can map any complex nonlinear relationship, making them suitable for handling nonlinear relationships between water quality indicators. Ouma et al. [14] presented an approach based on the feedforward neural network (FNN) model for the simulation and prediction of dissolved oxygen, and the research results indicate that feedforward backpropagation neural networks are effective methods for predicting dissolved oxygen. He et al. [15] proposed a water quality prediction method based on backpropagation neural network (BPNN), which utilizes an improved bee colony algorithm to optimize the neural network model and improve the accuracy of prediction. Since Long short-term memory network (LSTM) can make full use of the time series characteristics between monitoring data, more and more scholars use LSTM to solve the problem of water quality prediction. Hu et al. [16] proposed a new water quality prediction method based on deep learning network to predict pH and water temperature. In this method, Pearson correlation coefficient is used to obtain relevant knowledge between pH, water temperature, and other water quality parameters. However, this method requires more computational costs in dataset processing and the overall prediction results have strong fluctuations. Liu et al. [17] proposed a water quality prediction method based on LSTM deep neural network and established the sample, data pre-processing, parameter setting, and learning procedure of LSTM deep neural network. The established prediction model is suitable for automatic training and learning of different water quality data samples. Zhang et al. [18] proposed a watershed water quality prediction model based on attention mechanism and

Bidirectional LSTM (Bi-LSTM). The model mainly consists of a Bi-LSTM layer and a time attention layer, and introduces attention mechanism after bidirectional feature extraction of water quality time series data to highlight data sequences that have a critical impact on the prediction results. D. Ren et al. [19] proposed a hybrid model combining LSTM Seq2Seq with XGBoost and KNN, based on real-time conductivity (EC) and pH data from a water treatment plant in Korea. The model reduces RMSE by 12% and MAE by 50% in short-term (3-12 hours) and long-term (24-48 hours) predictions, respectively, significantly lowering the false alarm rate compared to traditional threshold methods, providing a reliable tool for dynamic water quality management. The summary of these water quality prediction methods are presented in Table 1.

III. OUR METHOD

This paper proposes a water quality prediction model that integrates CEEMDAN with a CNN-LSTM architecture, enhanced by an attention mechanism. This model is designed to overcome the limitations of traditional LSTM models in handling sequential data. The model effectively captures local features in time series data and reduces the dependency of LSTM on long sequence data. Moreover, it addresses the overfitting problem, enhancing the model's stability and improving its prediction accuracy.

A. CEEMDAN

CEEMDAN is an advanced signal processing technique designed to address the limitations of traditional Empirical Mode Decomposition (EMD) methods, particularly in handling noise interference [20]. The development of CEEMDAN arose from the need to mitigate the frequent noise-related challenges encountered in signal processing. Traditional EMD methods often suffer from instability and significant distortion in decomposition results when process-

ing noisy signals. To address these issues, CEEMDAN introduces a novel decomposition strategy that incorporates an adaptive noise adjustment mechanism, effectively reducing the impact of noise and enhancing both the accuracy and stability of the decomposition process. The decomposition steps of CEEMDAN are as follows:

Step 1 Add white noise to the time series $x(t)$ of water quality indicators, repeating this process n times with different white noise $\varepsilon_k \delta_i(t)$ each time, to generate a series of new noisy data sequences. The addition of white noise can effectively break the modal aliasing in the signal, helping to independently extract different frequency components of the signal, thereby improving the stability and accuracy.

$$x_i(t) = x(t) + \varepsilon_k \delta_i(t) \quad (1)$$

where ε_k the weight coefficient of Gaussian white noise, which changes with the decomposition levels, and $\delta_i(t)$ is the Gaussian white noise generated during the i -th processing. $x_i(t)$ represents the i -th newly generated sequence.

Step 2 Using the EDM method, decompose each newly generated sequence $x_i(t)$ to obtain its modal components and residual terms. Calculate the average of the first IMF component obtained from each decomposition, then determine the average value of the first IMF, and designate this as the first component $\overline{IMF}_1(t)$ of the decomposition result. Subtract this first component $\overline{IMF}_1(t)$ from the original sequence $x(t)$ to obtain the residual signal $r_1(t)$.

$$IMF_1(t) = \frac{1}{n} \sum_{i=1}^n E_1(x(t) + \varepsilon_0 E_1(\delta_i(t))) \quad (2)$$

$$r_1(t) = x(t) - IMF_1(t) \quad (3)$$

Where $IMF_1(t)$ is the average value of modal components; E_1 is an operator that performs a complete EMD on the input signal and extracts only the first IMF component.

Step 3 The residual term $r_1(t)$ is used as the new original sequence, and steps (1) and (2) are repeated to add different noise sequences to $r_1(t)$ to obtain a new noisy sequence $r_1(t) + \varepsilon_1 E_1(\delta_i(t))$. Then, EMD decomposition is performed on the new sequence to obtain the second IMF component.

$$IMF_2(t) = \frac{1}{n} \sum_{i=1}^n E_1(r_1(t) + \varepsilon_1 E_1(\delta_i(t))) \quad (4)$$

Step 4 Repeat the above steps, add Gaussian white noise to the residual, and perform EMD decomposition until the residual cannot be decomposed any further, to obtain the final residual.

$$r(t) = x(t) - \sum_{j=1}^n IMF_j(t) \quad (5)$$

Step 5 If the stopping condition of EMD is met, the iteration will be stopped, and the original signal is ultimately decomposed into k sets of modal components and the

remaining residual term $res(t)$. The final original sequence will be decomposed as shown below.

$$x(t) = \sum_{j=1}^k IMF_j(t) + res(t) \quad (6)$$

B. CNN

CNN consists of five main components: the input layer, convolutional layer, pooling layer, fully connected layer, and output layer [21], as illustrated in Figure 1.

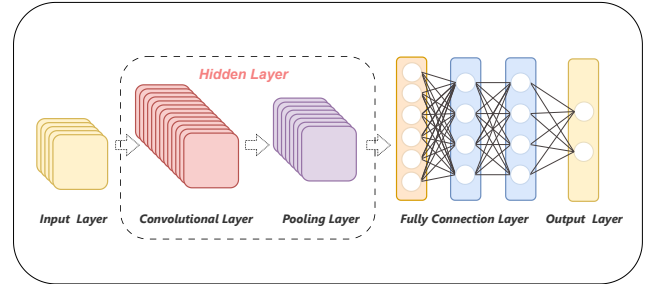


FIGURE 1: Convolutional Neural Network Architecture.

The convolutional layer is the core component of CNN, which extracts local features of input data through convolution operations. The calculation formula is as follows:

$$Y_{i,j}^l = \sigma \left(\sum_{c=1}^{C_{in}} \sum_{k=1}^K \omega_{c,j,k}^l X_{i+k-1,c}^{l-1} + B_j^l \right) \quad (7)$$

where $Y_{i,j}^l$ represents the output at position i the j -th channel in the l -th convolutional layer, C_{in} is the number of input channels, K is the kernel size, $\omega_{c,j,k}^l$ is the weight of the kernel for input channel c , output channel j , and kernel position k , B_j^l is the bias term, and σ is the activation function.

The calculation formula for the pooling layer is as follows:

$$P_{ij}^l = \max_{r \in R} (Y_{i \times T + r, j}^{l-1}) \quad (8)$$

where P_{ij}^l represents the output result of the pooling layer, T and R respectively represent the stride and kernel size of the pooling layer, and r is the pooling window.

By alternately applying convolutional and pooling layers, the model efficiently extracts local data features and reduces feature dimensionality. Furthermore, weight sharing not only minimizes the number of parameters but also decreases the model's complexity.

When dealing with time series data, CNNs treat the data as one-dimensional signals and employ one-dimensional convolution kernels to capture local features. These kernels slide along the time axis, identifying patterns and features at various time scales.

C. LSTM

LSTM is an advanced variant of Recurrent Neural Networks (RNNs) designed to address the issues of vanishing and exploding gradients during training, thereby enhancing the

model's ability to capture long-term dependencies [22]. The basic components of the LSTM model are illustrated in Figure 2. LSTM networks store and access information through fundamental memory cells and utilize gating mechanisms to regulate the flow of information. The key components of an LSTM network include the input gate, forget gate, and output gate. The input gate determines which data should be retained in the memory cell by combining the outputs of the sigmoid and tanh activation functions. The forget gate processes the input, previous state, and current memory cell state to decide which information should be discarded. The output gate controls the information to be output by determining the hidden state and applying the tanh activation function. The LSTM calculation formula is as follows:

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + B_i) \quad (9)$$

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + B_f) \quad (10)$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + B_o) \quad (11)$$

$$\tilde{c}_t = \tanh(W_{xc}x_t + W_{hc}h_{t-1} + B_c) \quad (12)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (13)$$

$$h_t = o_t \odot \tanh(c_t) \quad (14)$$

In the above formulas, i_t , f_t , o_t , c_t , $\tilde{c}_t$ represents the input gate, forget gate, output gate, memory unit state, and candidate memory unit state at time t , respectively. $\sigma(\cdot)$ represents the sigmoid activation function. W_{xi} , W_{xf} , W_{xo} and W_{xc} respectively represent the weight vectors of the input feature variables to i , f , o , and c . W_{hi} , W_{hf} , W_{ho} and W_{hc} respectively represent the weight vectors from the hidden layer to i , f , o and c . B_i , B_f , B_o , B_c respectively represent the offsets of i , f , o , and c . $\odot$ represents multiplying by elements.

D. SELF ATTENTION MECHANISM

The attention mechanism is a method in deep learning that simulates human attention allocation, enabling models to focus on the most important parts when processing data [23]. The core idea is to calculate the correlation between each element and all other elements to obtain a specific representation of each element. The basic structure of the self-attention mechanism includes Query, Key, and Value. These components are used, respectively, to calculate attention scores, represent elements, and produce results. The specific calculation methods are as follows.

(1) Calculate attention score: Use Query and Key to calculate the association between each element and all other elements. The input sequence is multiplied with the pre-trained weight matrices to generate a group of query vectors, key vectors, and value vectors. In this paper, the input sequence is the feature map output from the CNN layer:

$$Q = NW_q \quad (15)$$

$$K = NW_k \quad (16)$$

$$V = NW_v \quad (17)$$

$$Score_{i,j} = \frac{Q_i K_j^T}{\sqrt{d_k}} \quad (18)$$

where the weight matrices W_q , W_k and W_v are trainable parameter matrices initialized via glorot_uniform initialization. Q represents the query matrix, K represents the key matrix, and V represents the value matrix, N represents the feature map, $score_{i,j}$ is the attention score between the $i - th$ and $j - th$ elements, Q_i is the $i - th$ Query vector and K_j is the $j - th$ key vector, $\frac{1}{\sqrt{d_k}}$ the scaling factor, where d_k denotes the dimensionality of the key vectors. (2) Calculate attention weights: Convert attention scores into attention weights using the Softmax function. The attention weight between the $i - th$ element and all other elements can be expressed as:

$$W_{i,j} = \frac{\exp(Score_{i,j})}{\sum_{k=1}^n \exp(Score_{i,k})} \quad (19)$$

where $W_{i,j}$ represents the attention weight between the $i - th$ element and the $j - th$ element. n is the number of elements in the sequence.

(3) Calculate output result: Calculate the output result of each element based on Value and attention weight. The output result of the $i - th$ element can be expressed as:

$$O_i = \sum_{j=1}^n W_{i,j} V_j \quad (20)$$

where O_i is the attention output result of the $i - th$ element, V_j is the value of the $j - th$ element.

E. WATER QUALITY PREDICTION MODEL BASED ON CEEMDAN-CNN-LSTM

The proposed model consists of three modules: data processing, prediction model, and result evaluation. The framework is illustrated in Figure 2. The data processing module implements the CEEMDAN decomposition of water quality time-series data to obtain a set of IMF components, which represent different frequency components of the original data and are subsequently used as input for the prediction model. The water quality prediction module is a hybrid model that integrates the LSTM, CNN, and self-attention mechanism. Each IMF component derived from the data-processing module underwent feature extraction for long-term dependencies using an LSTM layer. The CNN layer then applies convolution to the LSTM output to capture the local features and generate a feature map. The self-attention layer assigns weights to the feature maps extracted by the CNN layer and computes the attention scores, thereby enhancing the focus of the model on the most salient features. The flattening layer subsequently processes the output of the self-attention layer to prepare the data for the input into the fully connected layer for the final prediction. The output of the water quality prediction module consisted of the prediction results for each IMF component. In the

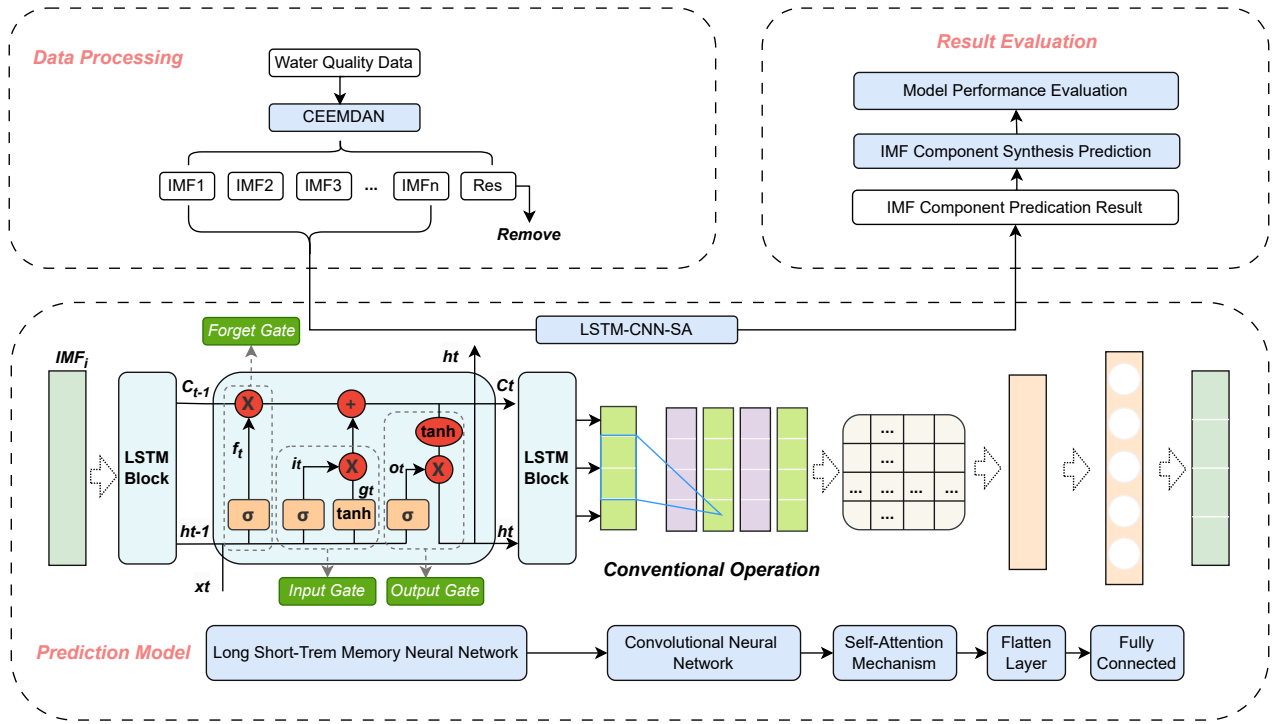


FIGURE 2: The framework of water quality prediction model.

result evaluation module, the prediction results for the individual IMF components were aggregated through a linear combination to derive the final prediction. The performance of the model was assessed using relevant evaluation metrics.

IV. EXPERIMENT

In this section, we conducted a series of experiments using real datasets to validate the effectiveness of the proposed method and compared its performance with some classic approaches.

A. EXPERIMENTAL SETUP

This experiment was conducted on a computer with AMD Ryzen 7 processor, 3.2GHz clock frequency, 16GB of memory, and 8 cores and 16 threads. The deep learning framework used was TensorFlow, using Python.

The experiment utilizes a real dataset from the Xixiyuan Reservoir Monitoring Station, covering the period from January 1, 2021, to April 18, 2023. The data is updated every three hours by the station, including indicators such as dissolved oxygen (DO), pH, chemical oxygen demand (CODMn), and turbidity (NTU). The dataset contains 6,999 valid entries. For the experiment, the first 5,600 entries were used as the training set, while the remaining 1,399 entries were reserved for the testing set.

In addressing the missing values within the dataset, we employed the K-Nearest Neighbors (KNN) algorithm for imputation. Given the significant differences in the value ranges of various water quality indicators, we aimed to

prevent the model from disproportionately focusing on features with larger value ranges, which could adversely affect both the convergence speed and overall performance of the model. Therefore, we performed normalization on the data, scaling it to the [0, 1] interval. The normalization calculation formula is as follows:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (21)$$

where X' is the normalized value, X is the original value, X_{min} is the minimum value of the feature, and X_{max} is the maximum value of the feature.

B. EVALUATION METRIC

In this study, the predictive performance of the model was evaluated based on the mean absolute error (MAE), root mean square error (RMSE), and coefficient of determination (R^2). The formulas for these metrics are as follows.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (22)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (23)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)}{\sum_{i=1}^n (y_i - \bar{y}_i)} \quad (24)$$

In the above formula, n represents the number of samples, y_i represents the actual value, $\hat{y}_i$ represents the predicted value, and $\bar{y}$ represents the average of the actual values.

C. EXPERIMENTAL RESULTS AND COMPARISON

Using the prediction of Dissolved Oxygen (DO) as an illustrative example, Figure 3 shows the results obtained from the CEEMDAN decomposition of water quality data. In the figure, the horizontal axis represents the date, whereas the vertical axis represents the amplitude. Following CEEMDAN decomposition, the data sequence yielded 13 components: 12 Intrinsic Mode Function (IMF) components and one residual (Res) component. The IMF components, labeled IMF1-IMF12, are arranged in descending order of frequency. The higher-frequency components exhibit rapid, erratic fluctuations with smaller amplitudes, reflecting short-term, random variations in dissolved oxygen content that lack clear patterns. Conversely, the lower-frequency components showed larger amplitudes, reflecting long-term trends and periodic fluctuations of DO in water quality.

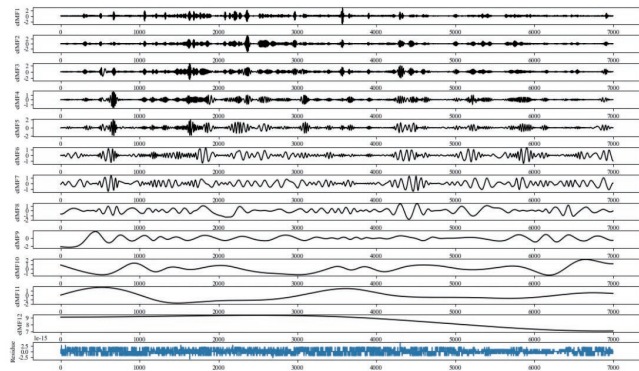


FIGURE 3: Decomposition Results of DO.

To demonstrate the superiority of the proposed model, we compared its prediction results with four baseline models: SVM [24], CNN-LSTM (CL) [25], and CEEMDAN-CNN-LSTM (CCL) [26], SELF-ATTENTION-CNN-LSTM (SACL) [27]. In this experiment, we set the prediction period to 3 hours. The experimental results are presented in Table 2 using dissolved oxygen as an illustrative example. A comparison of the prediction outcomes across the four models reveals that the SVM model performs the worst, primarily because of its inability to capture the intricate dynamics within time-series data. The CNN-LSTM model outperformed SVM in processing time-series data, indicating its superior capability. Furthermore, the CEEMDAN-CNN-LSTM model exhibited better performance than both CNN-LSTM and SVM, highlighting the positive contribution of the CEEMDAN algorithm to time-series prediction. The SELF-ATTENTION-CNN-LSTM model is an ablation experiment in which the CEEMDAN method is removed from the proposed model. The experiment demonstrated that the CEEMDAN decomposition method significantly

enhanced the performance of the water quality prediction model. After removing this method, the model's prediction accuracy clearly decreased, with RMSE and MAE increasing and R^2 decreasing, indicating that CEEMDAN helps to decompose the data into components with different frequencies, providing the model with more stable and clearer signals. Therefore, it plays an important role in improving the accuracy of the model by offering more precise input. By incorporating a self-attention mechanism into the CEEMDAN-CNN-LSTM model, our proposed model achieved superior predictive performance compared to the other four models, thereby validating the rationale behind our modeling approach. In the experiment, we iteratively tested various parameter values, observed the resulting changes, and identified the optimal parameters. The final parameter settings are listed in Table 3.

TABLE 2: Comparison results of DO prediction

Method	RMSE	MAE	R^2
SVM	0.463	0.378	0.926
CL	0.277	0.165	0.924
CCL	0.157	0.112	0.990
SACL	0.310	0.192	0.965
Proposed	0.138	0.084	0.993

TABLE 3: Parameter settings

Parameter	Value	Description
LOOK_BACK	10	Size of observation window
SPLIT	0.8	Proportion of dataset splitting
EPOCHS	100	Number of training rounds
BATCH_SIZE	100	The number of samples used in each training iteration
LAYER	128	The number of neurons in LSTM
FILTERS	350	The number of filters in the convolutional layer
KERNEL_SIZE	1	Convolutional kernel size in convolutional layers

To further validate the universality of the proposed model for multiparameter water quality prediction, prediction experiments were conducted on four key water quality indicators: turbidity (NTU), pH value (pH), dissolved oxygen (DO), and electrical conductivity (EC). The results are shown in Figure 4. The x-axis represents the timestamp with a 3-hour interval, and the y-axis represents the measured and predicted values for each parameter. The comparison reveals that the predicted curves, shown in blue, closely match the actual values, shown in yellow, for both overall trends and local fluctuations. For example, in pH prediction, the model accurately captures its relatively stable periodic variations; for the more volatile NTU parameter, the model effectively fits its short-term sudden increases or decreases. These results indicate that the CEEMDAN-LSTM-CNN model with self-attention is not only suitable for dissolved oxygen prediction but also generalizes effectively to water quality parameters with different dynamic characteristics.

TABLE 4: Comparison of different predication periods

Model	RMSE				MAE				R^2			
	6h	9h	12h	24h	6h	9h	12h	24h	6h	9h	12h	24h
SVM	0.528	0.575	0.620	0.732	0.427	0.462	0.498	0.586	0.897	0.878	0.858	0.802
CL	0.496	0.548	0.568	0.681	0.327	0.360	0.387	0.470	0.909	0.889	0.881	0.829
CCL	0.252	0.270	0.323	0.457	0.199	0.185	0.211	0.317	0.977	0.970	0.961	0.922
SACL	0.394	0.453	0.498	0.618	0.266	0.316	0.318	0.420	0.943	0.924	0.908	0.859
proposed	0.199	0.258	0.298	0.425	0.140	0.158	0.176	0.257	0.985	0.976	0.967	0.931

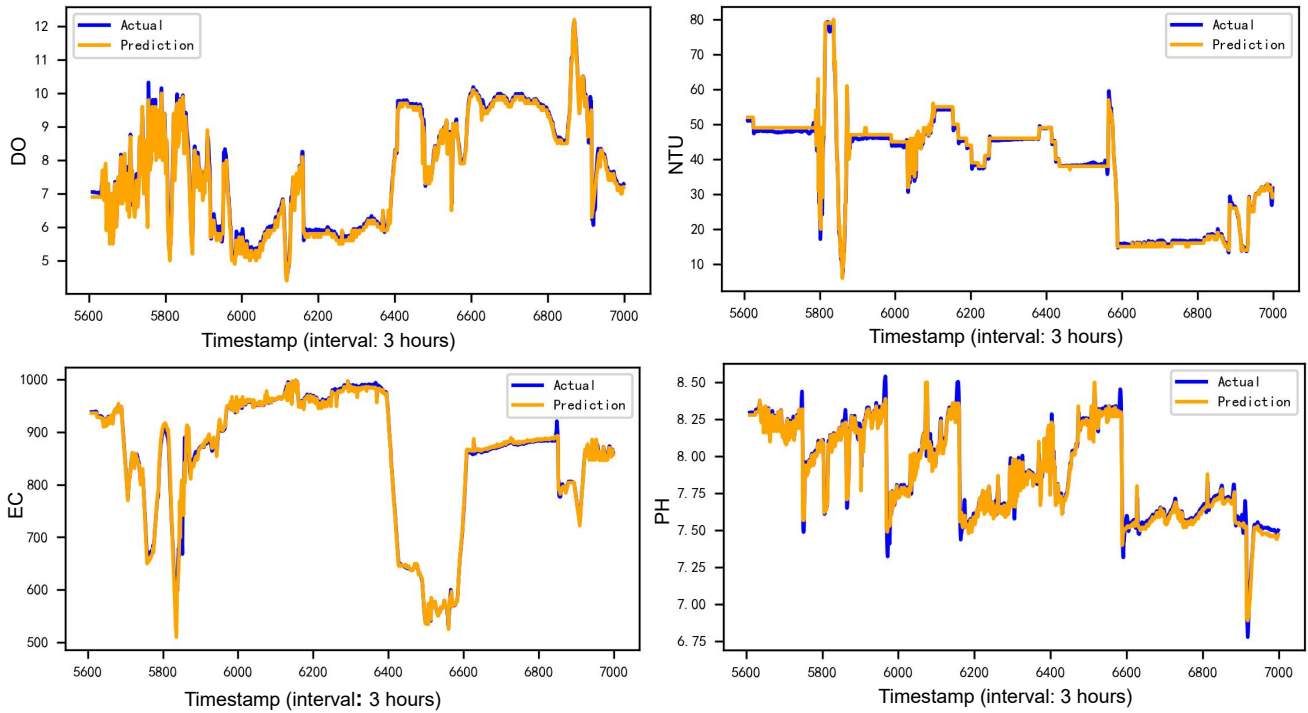


FIGURE 4: Performance of different indicators

This success is attributed to the adaptive processing capability of the CEEMDAN decomposition for nonstationary data, combined with the LSTM-CNN joint modeling of local features and long-term dependencies, as well as the dynamic weighting of key time steps through the self-attention mechanism. Furthermore, the robustness of the model in multiparameter prediction demonstrates its potential application value in multivariable water quality analysis, offering feasible technical references for the integrated management of complex water environments.

To comprehensively evaluate the model's predictive ability across different time scales, the prediction periods were set to 6, 9, 12, and 24 hours. The experimental results are shown in Figure 5. The blue curve represents the actual observed values, and the yellow curve represents the predicted values of the model. From the figure, it is evident that when the prediction period is relatively short, such as 6 hours, the predicted values closely match the actual values, indicating that the model can accurately capture the short-

term dynamic changes in water quality parameters. As the prediction period increased, the prediction error gradually accumulated, with deviations increased for some time steps when the prediction period reached 24 hours. Despite this, the overall predicted curve closely followed the actual trend. This result suggests that despite the increased uncertainty with longer-term predictions, the proposed model can dynamically adjust its attention to key historical time points through the self-attention mechanism, effectively mitigating error propagation and maintaining high prediction accuracy.

Table 4 presents the results of RMSE, MAE, and R^2 for the different models across various prediction cycles. The findings show that as the prediction period increases, the performance of all models deteriorates due to the accumulation of uncertainty in long-term predictions, amplifying prediction errors. The SVM model performs the worst due to its limited ability to capture time dependencies, especially over extended periods. In contrast, the combination of LSTM and CNN shows significant advantages, with LSTM

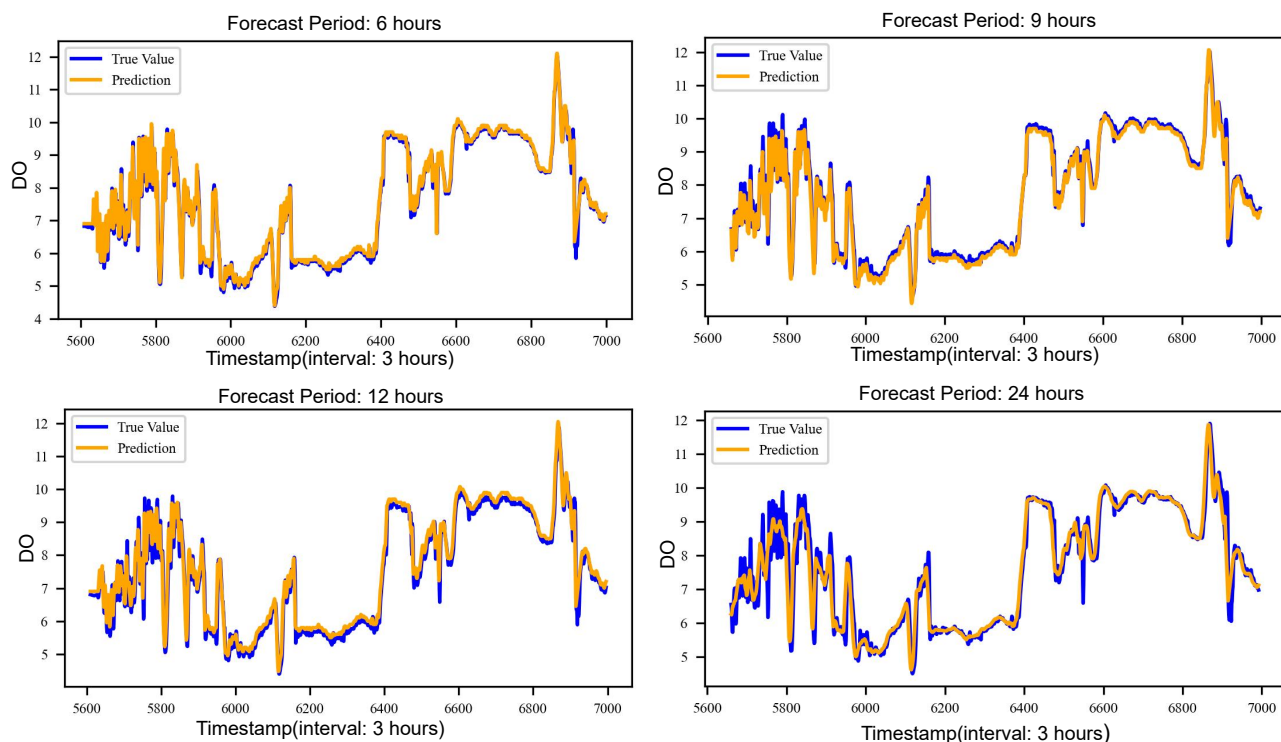


FIGURE 5: Performance of different predication periods.

capturing long-term dependencies and CNN excelling in extracting local features. The synergy of these models makes the LSTM-CNN model well-suited for long-term time-series prediction tasks. The CCL model outperforms both the SVM and CL models across all prediction periods. The performance of the SACL model is significantly lower than the CCL and Proposed models, indicating the positive impact of the CEEMDAN decomposition method on prediction accuracy. The proposed model excels across all prediction cycles, significantly outperforming the other models and maintaining high accuracy in long-term predictions. This advantage is due to the self-attention mechanism, which captures long-term dependencies and enhances the model's ability to use historical information, improving prediction accuracy. These results demonstrate that the proposed model offers superior stability and accuracy compared to traditional methods in long-term predictions.

V. CONCLUSIONS AND FUTURE WORK

This study proposes a water quality data prediction model based on hybrid machine learning with data preprocessing as a prerequisite. The CEEMDAN method was used to decompose and reconstruct the original data and help decompose the data into components with different frequencies, providing the model with more stable and clearer signals. Building on this foundation, a hybrid model incorporating CNN, LSTM, and a self-attention mechanism was employed to predict water quality data. Experimental validations show that the proposed model offers significant advantages over

traditional methods, as demonstrated by the improvements in the RMSE, MAE, and R^2 evaluation metrics. However, despite its excellent performance, this method has several limitations. First, the model relies solely on historical water quality data and does not incorporate the influence of external environmental factors on the water quality dynamics. For instance, rainfall events can significantly alter turbidity (NTU) through surface runoff, and such multi-source correlations have not been modeled. Second, the experimental data were derived from a single monitoring station, which overlooks the spatial heterogeneity of water bodies and limits the model's generalization capability. Third, the proposed hybrid model has a relatively complex structure, leaving room for further performance improvement. Therefore, future research should focus on addressing these challenges. Furthermore, exploring other advanced deep learning techniques for water quality prediction is another area of interest.

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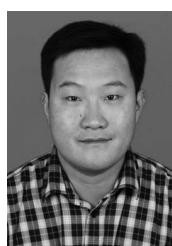
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